PersonaScienceBench

Same scientific task. Different collaborators.

STATIC EVALUATION
Task → Answer → Score

INTERACTIVE SCIENTIFIC EVALUATION
Agent ⇄ **Simulated researcher** → Scientific output + interaction trace

a Scientific task construction



Scientific task

Public inputs
Brief · data · tools

Researcher context
Study details
Requirements
Method guidance

b Persona-conditioned interaction

Interactive scientific collaboration
Approach + result review
or specification clarification

Persona
MatrAIx Profile
Communication & style

Scientific agent

Ask · propose
report results

Clarify · review
request revisions

Researcher simulator

Researcher-held scientific context

c Evaluation

Submission

Scientific outcome
Grader
Private references
Scientific correctness

Collaboration Adaptation
Information recovery
Dialogue & adaptation
Revisions & repetition
Tokens & tool calls